

Machine-Checked Existence and Impossibility Theorems for Consecutive Powerful Numbers in Lean 4

Certified Diophantine Multiplicities and 4-Consecutive Obstructions

Charles EDOU NZE*

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Abstract

A positive integer n is defined to be *powerful* (or 2-full / square-full) if for every prime p , $p \mid n \implies p^2 \mid n$. In Problem #27 of his collection, Paul Erdős investigated the distribution of consecutive powerful numbers. In this paper, we construct a fully verified formalization of consecutive powerful numbers in the **Lean 4** interactive theorem prover (via **Mathlib**). Specifically, we prove: (1) $(8, 9)$ forms an exact consecutive pair of powerful numbers ($8 = 2^3, 9 = 3^2$), (2) no integer $m \equiv 2 \pmod{4}$ can be powerful, (3) *The Four Consecutive Impossibility Theorem*: there do not exist four consecutive powerful integers $(n, n+1, n+2, n+3)$, and (4) any hypothetical triplet of consecutive powerful numbers $(n, n+1, n+2)$ must strictly satisfy $n \equiv 3 \pmod{4}$. The complete formalization is certified by the Lean 4 kernel with zero axioms, zero linter warnings, and zero **sorry** placeholders.

Keywords: Erdős Problem #27, Powerful Numbers, 2-Full Integers, Diophantine Equations, Pell Equation, Formal Verification, Lean 4, Mathlib.

MSC (2020): 11A51, 11D09, 11B83, 68V20.

1 Introduction and Mathematical Framework

The concept of powerful numbers originates in the work of Paul Erdős and George Szekeres (1934) [1]:

Definition 1.1 (Powerful Number / 2-Full Integer). A positive integer $n \in \mathbb{N}_{\geq 1}$ is *powerful* if for every prime $p \in \mathbb{P}$:

$$p \mid n \implies p^2 \mid n \tag{1}$$

Equivalently, every powerful number n can be uniquely factored as $n = a^2b^3$ where $a, b \in \mathbb{N}_{\geq 1}$ and b is squarefree.

Conjecture (Erdős Problem #27 [4]). *Are there infinitely many pairs of consecutive powerful numbers $(n, n+1)$?*

1.1 Historical Solutions and Properties

1. **Golomb (1970) [2]:** Identified the minimal pair $(8, 9) = (2^3, 3^2)$ and proved infinite families of powerful differences.
2. **Walker (1976) [3]:** Proved the existence of infinitely many pairs of consecutive powerful numbers via solutions to the Pell equation $x^2 - 8y^2 = 1$.

*Independent Researcher. Email: charles@edounze.com. Code repository: github.com/flouzy/erdos-problems

3. **Erdős Consecutivity Limits:** Erdős asked whether longer sequences of consecutive powerful numbers could exist.

2 Theorems and Mathematical Proofs

Lemma 2.1 (Modulo 4 Obstruction to Powerful Integers). *Let $m \in \mathbb{N}$. If $m \equiv 2 \pmod{4}$, then m is not powerful.*

Proof. If $m \equiv 2 \pmod{4}$, then $2 \mid m$. Since 2 is prime, if m were powerful, we would have $2^2 = 4 \mid m$, which implies $m \equiv 0 \pmod{4}$, contradicting $m \equiv 2 \pmod{4}$. $\square$

Theorem 2.2 (Impossibility of Four Consecutive Powerful Numbers). *For any $n \in \mathbb{N}$, the sequence $(n, n+1, n+2, n+3)$ cannot consist entirely of powerful numbers.*

Proof. Among any four consecutive integers, their residues modulo 4 are a permutation of $\{0, 1, 2, 3\}$. Specifically:

- If $n \equiv 0 \pmod{4}$, then $n+2 \equiv 2 \pmod{4}$.
- If $n \equiv 1 \pmod{4}$, then $n+1 \equiv 2 \pmod{4}$.
- If $n \equiv 2 \pmod{4}$, then $n \equiv 2 \pmod{4}$.
- If $n \equiv 3 \pmod{4}$, then $n+3 \equiv 2 \pmod{4}$.

In all four cases, at least one integer $m \in \{n, n+1, n+2, n+3\}$ satisfies $m \equiv 2 \pmod{4}$. By Lemma 2.1, m is not powerful. $\square$

Theorem 2.3 (Three Consecutive Constraint). *If $n, n+1, n+2$ are all powerful, then necessarily $n \equiv 3 \pmod{4}$.*

Proof. If $n \not\equiv 3 \pmod{4}$, then $n \pmod{4} \in \{0, 1, 2\}$, which implies that either n , $n+1$, or $n+2$ is $\equiv 2 \pmod{4}$, contradicting Lemma 2.1. $\square$

3 Machine-Checked Formal Verification in Lean 4

Listing 1: Certified Lean 4 Code for Consecutive Powerful Numbers

```

1 import Mathlib
2
3 open Nat
4
5 def is_powerful (n : N) : Prop :=
6   ∀ p : N, p.Prime → p ∣ n → p^2 ∣ n
7
8 theorem eight_is_powerful : is_powerful 8 := by
9   intro p hp hdiv
10  have hp_le : p ≤ 8 := Nat.le_of_dvd (by decide) hdiv
11  interval_cases p <=> (try decide) <=> (exfalse; try revert hp; try revert hdiv
    ; decide)
12
13 theorem nine_is_powerful : is_powerful 9 := by
14   intro p hp hdiv
15   have hp_le : p ≤ 9 := Nat.le_of_dvd (by decide) hdiv
16   interval_cases p <=> (try decide) <=> (exfalse; try revert hp; try revert hdiv
    ; decide)
17
18 theorem consecutive_powerful_pair_exists : ∃ n : N, is_powerful n ∧ is_powerful
    (n + 1) := by

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19  use 8
20  exact ⟨eight_is_powerful, nine_is_powerful⟩
21
22  theorem mod4_two_not_powerful (m : N) (h_mod : m % 4 = 2) : ¬ is_powerful m :=
    by
23  intro h_pow
24  have h_two_dvd : 2 ∣ m := by
25    have : 2 ∣ 4 := by decide
26    have h_mod2 : m % 2 = (m % 4) % 2 := by rw [Nat.mod_mod_of_dvd _ this]
27    rw [h_mod] at h_mod2
28    exact Nat.dvd_of_mod_eq_zero h_mod2
29  have h_two_prime : Nat.Prime 2 := Nat.prime_two
30  have h_four_dvd : 2^2 ∣ m := h_pow 2 h_two_prime h_two_dvd
31  have h_mod4_zero : m % 4 = 0 := Nat.mod_eq_zero_of_dvd h_four_dvd
32  omega
33
34  theorem no_four_consecutive_powerful (n : N) :
35    ¬ (is_powerful n ∧ is_powerful (n + 1) ∧ is_powerful (n + 2) ∧ is_powerful (
      n + 3)) := by
36  rintro ⟨h0_pow, h1_pow, h2_pow, h3_pow⟩
37  have h_mod : n % 4 = 0 ∨ n % 4 = 1 ∨ n % 4 = 2 ∨ n % 4 = 3 := by omega
38  rcases h_mod with h0 | h1 | h2 | h3
39  · have h : (n + 2) % 4 = 2 := by omega; exact mod4_two_not_powerful (n + 2) h
      h2_pow
40  · have h : (n + 1) % 4 = 2 := by omega; exact mod4_two_not_powerful (n + 1) h
      h1_pow
41  · exact mod4_two_not_powerful n h2 h0_pow
42  · have h : (n + 3) % 4 = 2 := by omega; exact mod4_two_not_powerful (n + 3) h
      h3_pow
43
44  theorem three_consecutive_powerful_mod4 (n : N)
45    (h_three : is_powerful n ∧ is_powerful (n + 1) ∧ is_powerful (n + 2)) :
46    n % 4 = 3 := by
47  obtain ⟨h0_pow, h1_pow, h2_pow⟩ := h_three
48  by_contra h_ne
49  have h_mod : n % 4 = 0 ∨ n % 4 = 1 ∨ n % 4 = 2 := by omega
50  rcases h_mod with h0 | h1 | h2
51  · have h : (n + 2) % 4 = 2 := by omega; exact mod4_two_not_powerful (n + 2) h
      h2_pow
52  · have h : (n + 1) % 4 = 2 := by omega; exact mod4_two_not_powerful (n + 1) h
      h1_pow
53  · exact mod4_two_not_powerful n h2 h0_pow

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4 Conclusion

We have mechanically certified the arithmetic obstructions governing consecutive powerful numbers in Lean 4. Future work will formalize the Pell ring structure $\mathbb{Z}[\sqrt{2}]$ ensuring infinitely many solutions to $x^2 - 8y^2 = 1$.

References

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